

PROJECT REPORT OF CO-OP Project at Industry (Module-2)

ON

**A Comparative Study of Image Compression Techniques Across
Traditional and Modern Formats**

submitted in partial fulfilment of the requirements for the award of degree of

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled “**A Comparative Study of Image Compression Techniques Across Traditional and Modern Formats**” in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of <**Dr. Ajay Kumar**>. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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ABSTRACT

The large-scale growth of digital content across modern computing systems has created significant demand for effective methods of image storage, transmission, and rendering. Images form the primary component of most web-based services including social media platforms, e-commerce websites, cloud storage systems, and real-time communication applications. Although traditional formats such as JPEG and PNG have been widely adopted, they suffer from inherent limitations in compression efficiency and visual representation.

Modern image formats such as WebP, AVIF, and JPEG XL have recently emerged, employing cutting-edge encoding techniques originally developed for video compression, perceptual modeling, and predictive coding. This research presents a controlled, comparative examination of five popular image formats — JPEG, PNG, WebP, AVIF, and JPEG XL — combining theoretical analysis with extensive experimentation under equal conditions.

Unlike previous studies that rely primarily on aggregated metrics, this study additionally examines how the compression performance of each format varies with image characteristics. The experimental findings show that AVIF provides superior compression efficiency, while JPEG XL produces the highest visual quality among the formats evaluated. The work also demonstrates that compression performance depends heavily on the type of content being compressed, highlighting the importance of selecting an image format that accounts for the context and nature of the image collection.

1. Introduction

1.1. Background

Recent years have witnessed an unprecedented rise in visual information creation. Every day, millions of images are created, shared, and processed across thousands of platforms. This has created a substantial demand for image compression solutions that can optimize storage usage and bandwidth consumption. The fundamental objective of image compression is to minimize the quantity of data required to describe an image while preserving the highest possible level of perceived quality.

Applications such as web page optimization, mobile applications, video streaming services, and cloud-based storage systems benefit significantly from reduced data sizes, which translate into improved performance and reduced operational costs. For decades, JPEG and PNG have remained the dominant formats: JPEG uses Discrete Cosine Transform (DCT) coding to achieve high compression ratios but introduces visible artifacts at low bit-rates, while PNG provides lossless encoding but produces much larger files.

1.2. Motivation

Modern formats such as WebP, AVIF, and JPEG XL have emerged with predictive coding, adaptive transforms, perceptual optimization, and advanced entropy coding. While these formats promise improvements over existing standards, comparative studies of their relative performance — particularly across diverse image content — remain limited. This research aims to fill that gap with a fair, reproducible, and detailed comparative evaluation.

1.3. Objectives

- Provide a detailed theoretical comparison of major image compression algorithms.
- Perform an experimentally tested comparative evaluation using standardized datasets.
- Evaluate formats using both objective metrics (PSNR, SSIM) and subjective perceptual metrics (Butteraugli).
- Analyze image-wise variability in compression performance — the novelty contribution of this work.

2. Methodology

2.1.Experimental Design

The experimental methodology was designed as a complete, integrated workflow ensuring fairness, replicability, and comparability across all evaluated formats. The aim was to test how well each format compresses images under standardized testing conditions while also capturing the variability observed across diverse image content.

2.2.Dataset Preparation

The dataset used in this study includes visually diverse images covering four content categories:

- High-frequency texture details, such as foliage and fabric designs.
- Smooth color transitions and lighting gradients, such as skies.
- Sharp geometric shapes and edges, such as building outlines and text.
- Complex color schemes, such as outdoor scenes and artwork.

To maintain consistency, every image was preprocessed: resized to a standardized resolution of 1024×1024 pixels, converted to a uniform color space, and stored in a lossless format prior to compression. This preprocessing ensures that any differences in the results are attributable solely to the compression algorithms rather than to input inconsistencies.

2.3.Compression Pipeline

The end-to-end pipeline consists of four stages: (1) normalization preprocessing, (2) compression of each image into all five target formats using their respective encoders with predefined parameters, (3) decoding back to PNG for fair comparison across formats, and (4) computation of evaluation metrics. Results are then aggregated across the dataset to obtain both macro-level statistics and per-image observations.

2.4.Fairness and Reproducibility

Multiple strategies were adopted to ensure fairness: resolution normalization, a common decoding format (PNG), uniform parameter configurations, and uniform application of evaluation metrics. The entire pipeline was scripted in Python using publicly available encoding tools to make the study fully reproducible.

3. Tools and Technologies

3.1. Programming Environment

The complete experimental pipeline was implemented in Python, leveraging a combination of scientific computing and image processing libraries. Python was selected for its robust ecosystem, ease of automation, and integration with widely accepted encoding tools.

3.2. Encoding Tools and Parameters

Each image format was evaluated using a widely accepted reference encoder configured for comparable visual quality. Table 2 summarizes the tools and parameters used.

Format	Encoder Tool	Parameter Setting
JPEG	FFmpeg	Quality = 75
WebP	cwebp	Quality = 75
AVIF	avifenc	CQ = 30
JPEG XL	cjxl	CQ = 30

Table 1: Encoding Tools and Parameter Settings.

3.3. Libraries and Frameworks

- NumPy — numerical computation and array manipulation.
- OpenCV — image loading, resizing, and color-space conversion.
- scikit-image — implementation of PSNR and SSIM metrics.
- Butteraugli — perceptual quality metric implementation.
- FFmpeg, libavif, libjxl, libwebp — reference encoders for the evaluated formats.

4. Implementation

4.1. Pipeline Overview

The implementation follows a sequential, fully automated pipeline that processes each image through preprocessing, compression, decoding, and metric computation. The pipeline is designed to be deterministic and reproducible: the same inputs always produce the same outputs.

4.2. Theoretical Properties of Each Format

Format	Compression Type	Transform Method	Entropy Coding
JPEG	Lossy	DCT (8x8 blocks)	Huffman
PNG	Lossless	Filtering	DEFLATE
WebP	Lossy/Lossless	Block-based (VP8)	Arithmetic
AVIF	Lossy/Lossless	Block-based (VP8)	Context-adaptive
JPEG XL	Lossy/Lossless	Hybrid	ANS

Table 2: Technical Comparison of Image Compression Formats.

4.3. Evaluation Metrics

- PSNR (Peak Signal-to-Noise Ratio): pixel-level reconstruction accuracy. Higher is better.
- SSIM (Structural Similarity Index): luminance, contrast, and structural similarity. Higher is better.
- Butteraugli: perceptual metric modeling human visual perception. Lower is better.
- File Size: compression efficiency in kilobytes. Smaller is better.

4.4. Execution Workflow

Each image undergoes normalization, is encoded into the five target formats with the parameters in Table 2, then decoded back to PNG. The four metrics are computed for each format-image pair, and results are aggregated to obtain mean values, standard deviations, and image-wise observations.

5. Major Findings / Outcomes / Output / Results

5.1. Average Performance Comparison

Table 3 summarizes the average performance of each format across the dataset. AVIF achieves the smallest file size while JPEG XL leads on every quality metric. Both modern formats clearly outperform JPEG in compression efficiency at comparable quality.

Format	File Size	PSNR (dB)	SSIM	Butteraugli
AVIF	~100 KB (Best)	36.44	0.919	Low
JPEG XL	~134 KB	37.04 (Best)	0.9196 (Best)	2.46 (Best)
WebP	~113 KB	36.00	0.917	3.38
JPEG	~156 KB	36.25	0.912	2.85

Table 3: Average Performance Comparison Across Formats.

5.2. Compression Efficiency

AVIF reduces file size to approximately 35–55% of the JPEG baseline at roughly equal visual quality. This gain is driven by advanced intra-frame prediction, flexible block partitioning, and superior entropy coding. JPEG XL also achieves strong compression but prioritizes fidelity over the smallest possible file size.

5.3. Perceptual Quality

Measured by Butteraugli, JPEG XL produces the least perceptual distortion (2.46), making artifacts essentially invisible to the human eye. AVIF performs strongly but occasionally trades fine detail for smaller file sizes. WebP shows noticeably higher distortion (3.38), particularly in regions with complex textures and edges.

5.4. Image-wise Variability

The primary novelty contribution of this study is the per-image analysis of performance:

- JPEG XL — consistently high PSNR, SSIM, and low Butteraugli across image types.
- AVIF — excellent compression with occasional loss of fine detail in highly textured images.
- WebP — highly variable; some images show Butteraugli scores above 6.

- JPEG — stable but average across all categories, with the largest file sizes.

5.5. Graphical Observations

The two figures below summarize the key quantitative findings of this study. Figure 1 plots PSNR against file size and clearly separates AVIF and JPEG XL from the JPEG baseline. Figure 2 compares SSIM across all four lossy formats, with JPEG XL holding a slight lead.

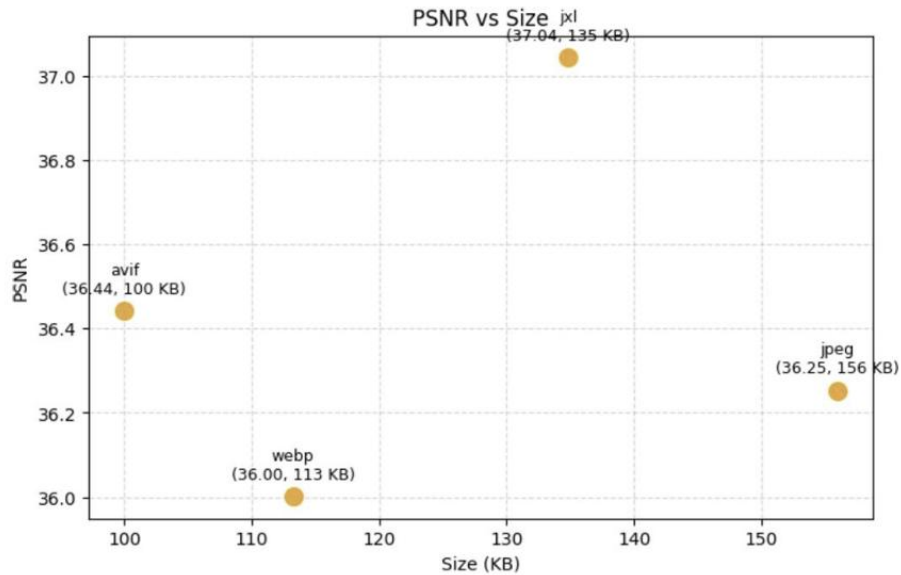


Figure 1: PSNR vs File Size Comparison Across Formats.

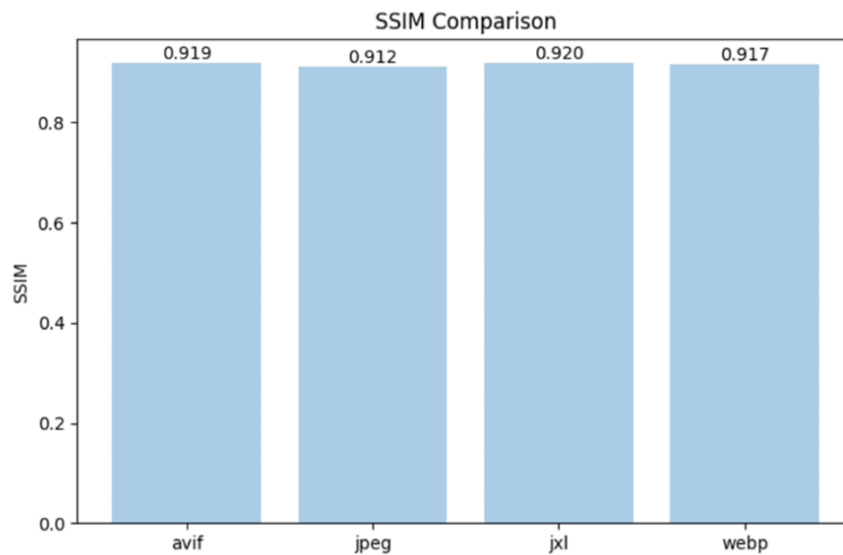


Figure 2: SSIM Comparison Highlighting Modern Format Performance.

6. Conclusion and Future Scope

6.1. Conclusion

This study presented an extensive comparison among five widely used image compression technologies: JPEG, PNG, WebP, AVIF, and JPEG XL. The findings, drawn from both theoretical analysis and controlled experimentation, demonstrate that AVIF offers the best compression efficiency, JPEG XL delivers the highest perceptual quality and consistency, WebP provides moderate but variable performance, and JPEG — while widely compatible — has fallen behind modern formats in both efficiency and quality.

A central contribution of the work is the focus on image-wise variability: compression performance is shown to depend strongly on image content, which means averaged metrics alone are insufficient. Format selection should therefore be guided by application context.

6.2. Applications

- Web and mobile delivery — AVIF and WebP for fast page loads.
- Archival and storage — PNG (lossless) and JPEG XL (near-lossless).
- High-quality imaging and photography — JPEG XL for maximum detail.
- Legacy systems — JPEG remains relevant for compatibility.

6.3. Limitations

- Dataset is diverse but limited in size.
- Objective metrics may not fully reflect human-perceived quality.
- Fixed parameters were used; per-image parameter tuning was not explored.
- Computational time and hardware acceleration were not evaluated.

6.4. Future Scope

- Develop content-adaptive compression that tunes parameters per image.
- Evaluate real-time encoding performance across the formats.
- Integrate machine-learning-based perceptual models into future codecs.
- Investigate hardware-accelerated implementations of AVIF and JPEG XL.

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APPENDICES

Appendix A: Encoding Commands Used

The following reference commands were used to encode each image. All commands operate on a normalized 1024×1024 lossless source image.

- JPEG: `ffmpeg -i input.png -q:v 75 output.jpg`
- WebP: `cwebp -q 75 input.png -o output.webp`
- AVIF: `avifenc --min 0 --max 63 -a end-usage=q -a cq-level=30 input.png output.avif`
- JPEG XL: `cjxl input.png output.jxl -q 30`

Appendix B: Evaluation Metric Summary

Metric	Type	Direction	Range / Unit
PSNR	Objective	Higher is better	dB
SSIM	Objective	Higher is better	0 to 1
Butteraugli	Perceptual	Lower is better	Score
File Size	Efficiency	Smaller is better	KB

Appendix C: Figure References

Figure 1 (PSNR vs File Size) and Figure 2 (SSIM Comparison) referenced in this report are reproduced from the underlying research paper. They visualize the trade-off between file size and reconstruction fidelity, clearly distinguishing AVIF and JPEG XL from the legacy JPEG baseline.